

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	9.4	4.6	2.3	0.9	0.3	82.5
C2 alpha=0 one-stage	9	5.4	2.6	0.9	0.6	81.5
C3 alpha=0 one-stage	9.1	5.5	3.1	0.9	0.4	81
Oracle C1 alpha=0 two-stage	9.2	5.1	2.2	0.8	0.3	82.4
C2 alpha=0 two-stage	8.9	6	3	0.8	0.4	80.9
C3 alpha=0 two-stage	9.3	6.3	3.2	0.9	0.4	79.9
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	11	4.2	2.3	0.9	0.2	
C2 alpha=0 one-stage	11	4.3	2.4	1	0.3	
C3 alpha=0 one-stage	10.9	4.4	2.5	1	0.3	
Oracle C1 alpha=0 two-stage	10.9	4.3	2.4	0.9	0.3	
C2 alpha=0 two-stage	10.9	4.4	2.5	1	0.3	
C3 alpha=0 two-stage	10.8	4.5	2.5	1	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	23.9	21	19.3	21.8	13.1	0.9
C2 alpha=0 one-stage	22.8	22	19.1	21.3	13.8	1
C3 alpha=0 one-stage	23.5	21.8	18.8	21.5	13.5	0.9
Oracle C1 alpha=0 two-stage	24.1	21.9	18.3	21.1	13.7	0.9
C2 alpha=0 two-stage	26.1	19.8	18	20.7	14.4	1
C3 alpha=0 two-stage	24.2	21.2	17.6	21.3	14.8	0.9
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.2	6.7	5.8	5.8	4.5	
C2 alpha=0 one-stage	7.2	6.7	5.8	5.7	4.7	
C3 alpha=0 one-stage	7.3	6.6	5.6	5.8	4.7	
Oracle C1 alpha=0 two-stage	7.1	6.7	5.7	5.8	4.8	
C2 alpha=0 two-stage	7.1	6.5	5.7	5.7	4.9	
C3 alpha=0 two-stage	7.1	6.6	5.6	5.8	4.8	

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	4.9	47	28.4	12.3	2.2	5.2
C2 alpha=0 one-stage	5.3	45	29.7	12.8	2.2	5
C3 alpha=0 one-stage	5.4	44.9	29.8	12.4	1.8	5.7
Oracle C1 alpha=0 two-stage	4.5	48.5	29.1	10.9	1.9	5.1
C2 alpha=0 two-stage	5.2	46.7	29.1	12.1	2.2	4.7
C3 alpha=0 two-stage	6	46.8	28.4	12.3	2.1	4.4
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	5.2	11.3	7.6	4.1	1.3	
C2 alpha=0 one-stage	5.2	10.9	7.7	4.3	1.4	
C3 alpha=0 one-stage	5.2	10.9	7.6	4.4	1.5	
Oracle C1 alpha=0 two-stage	5.1	10.7	8	4.3	1.4	
C2 alpha=0 two-stage	5	10.4	7.9	4.6	1.6	
C3 alpha=0 two-stage	5	10.4	7.7	4.8	1.7	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	0.8	1.4	6	59.4	32.4	0
C2 alpha=0 one-stage	1.1	1.4	4.7	59.6	33.2	0
C3 alpha=0 one-stage	1.2	1.1	4.9	59.7	33.1	0
Oracle C1 alpha=0 two-stage	1.1	1.2	6.3	56	35.4	0
C2 alpha=0 two-stage	1.5	1.4	5.4	55.9	35.8	0
C3 alpha=0 two-stage	1.2	1.3	4.5	55.9	37.1	0
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	3.4	3.5	4.5	12.5	6.2	
C2 alpha=0 one-stage	3.4	3.5	4.4	12.2	6.5	
C3 alpha=0 one-stage	3.4	3.5	4.3	12.4	6.4	
Oracle C1 alpha=0 two-stage	3.4	3.4	4.3	10.9	8	
C2 alpha=0 two-stage	3.4	3.4	4.3	10.7	8.2	
C3 alpha=0 two-stage	3.4	3.5	4.2	10.8	8.1	

Phase I/II Operating Characteristics

Continuous enrollment | 56-day arrival | IPDE alpha = 0

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	No OBD %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	10.7	25.3	44.6	9.5	0.6	9.3
C2 alpha=0 one-stage	10.9	21.9	46.5	12	0.7	8
C3 alpha=0 one-stage	11	22.8	46.2	11.4	0.4	8.2
Oracle C1 alpha=0 two-stage	11.3	23.8	46.3	9	0.9	8.7
C2 alpha=0 two-stage	10.8	23.3	45.9	11.2	1.1	7.7
C3 alpha=0 two-stage	11	22.9	44.7	11.9	1.6	7.9
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	7.3	8	9.8	3.4	0.6	
C2 alpha=0 one-stage	7.1	7.8	9.9	3.7	0.6	
C3 alpha=0 one-stage	7.1	7.8	10	3.7	0.7	
Oracle C1 alpha=0 two-stage	6.9	7.9	9.7	3.8	0.9	
C2 alpha=0 two-stage	6.8	7.5	9.7	4.1	1.1	
C3 alpha=0 two-stage	6.9	7.5	9.6	4.1	1.1	
Scenario 38						
DLT rate	0.05	0.1	0.12	0.15	0.3	
Efficacy rate	0.05	0.08	0.12	0.15	0.5	
OBD selection (%)						
Oracle C1 alpha=0 one-stage	3	6.5	8.7	11.8	65.3	4.7
C2 alpha=0 one-stage	3.5	5.8	8.5	12.1	65.6	4.5
C3 alpha=0 one-stage	3.7	6.1	7.1	12.4	66.5	4.2
Oracle C1 alpha=0 two-stage	4.6	7.2	8.6	11.6	63.2	4.8
C2 alpha=0 two-stage	4.7	7.3	8	12	64.2	3.8
C3 alpha=0 two-stage	4.4	7.1	8	11.6	64.9	4
Mean administrations by dose						
Oracle C1 alpha=0 one-stage	4.6	4.6	5	5.5	10.2	
C2 alpha=0 one-stage	4.5	4.4	5	5.5	10.5	
C3 alpha=0 one-stage	4.5	4.4	4.9	5.5	10.5	
Oracle C1 alpha=0 two-stage	4.5	4.5	4.9	5.6	10.3	
C2 alpha=0 two-stage	4.5	4.4	4.9	5.5	10.7	
C3 alpha=0 two-stage	4.5	4.4	4.9	5.4	10.7	

Mean Unique Patients by Method

Mean number of unique patients per simulated trial

Scenario	1	16	20	24	27	38	Mean
Oracle C1 alpha=0 one-stage	18.7	30	29.5	30	29.1	29.8	27.8
C2 alpha=0 one-stage	12.3	18.2	18.9	19.5	19.4	18.9	17.9
C3 alpha=0 one-stage	10.5	14.6	15.8	16.5	16.6	15.6	14.9
Oracle C1 alpha=0 two-stage	18.8	30	29.5	30	29.1	29.8	27.9
C2 alpha=0 two-stage	12.4	18.2	19	19.7	19.6	18.9	18
C3 alpha=0 two-stage	10.6	14.7	15.9	16.7	16.9	15.6	15.1

Values are reported from Total_Unique_Patients; comparator counts without patient IDs are sums of dose-level means.

Rows are one-stage first, then two-stage. C1 is fixed at alpha=0; C2/C3 use alpha=0.